Maulana Azad National Institute of Technology Bhopal

Department of Mathematics Bioinformatics and Computer Applications

End Term, May-2025

Programme	B.Tech	Semester	II
Course code	MTH24102	Course Title	Mathematics-II
Total Marks	40	Time	2 Hours

Answer all questions.

Q. No.	Question Text	Marks	CO	BT
1	(a) Solve the equations $\begin{cases} x_1 - x_2 + x_3 + x_4 = 2, \\ x_1 + x_2 - x_3 + x_4 = -4, \\ x_1 + x_2 + x_3 - x_4 = 4, \\ x_1 + x_2 + x_3 + x_4 = 0, \end{cases}$ applying Gauss-Jordan elimination method. (b) Find the values of λ and μ for which the system of equations $2x + 3y + 5z = 9$, $7x + 3y - 2z = 8$, and $2x + 3y + \lambda z = \mu$ has infinite number of solutions.	4+2	1	3
2	(a) Using the Laplace transformation, solve $\frac{d^2x}{dt^2} + 9x = \cos 2t$, if $x(0) = 1$, and $x(\pi/2) = -1$. (b) Solve the equation $f(t) = \cos t + \int_0^t e^{-u} f(t-u)\, du$, using Laplace transformation.	4+4	3	3
3	(a) Find the Fourier series expansion of $f(x) = x - x^2$ in $(-\pi, \pi)$. Deduce that $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots = \frac{\pi^2}{12}$. (b) Prove that the Fourier transformation of $e^{-\frac{x^2}{2}}$ is self-reciprocal. (Self-reciprocal means the Fourier transformation of f is f itself.)	5+4	4	3
4	(a) Solve the partial differential equation $2xz - px^2 - 2qxy + pq = 0$, where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$. (b) Find the general solution of the partial differential equation $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - \frac{\partial^2 z}{\partial y^2} = y \cos x$.	4+4	5	5
5	(a) Using the method of separation of variables, solve $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u$, where $u(x, 0) = 6e^{-3x}$. (b) Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ with initial condition $u(x, 0) = 3 \sin n\pi x$ and boundary conditions $u(0, t) = 0$, $u(1, t) = 0$, where $0 < x < 1$, $t > 0$.	4+5	5	3